Yitian Yu

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Objective

Ambitious and detail-oriented Computer Engineering student at Georgia Tech with a strong foundation in deep learning, data analytics, programming, and open-source collaboration. Looking for an opportunity to contribute to ongoing research operations and gain hands-on experience in a lab setting.

Education

Georgia Institute of Technology | Atlanta, GA *January 2017 – Present*
Bachelor of Science in Computer Engineering, GPA 3.56 Expected Graduation, May 2020

Skills

Programming: Java, Python (novice), C, C#, Ruby, JavaScript, PHP, SQL, Objective-C, Perl, Node.js, Go
Platforms: Linux
Hardware: FPGAs, oscilloscope, logic analyzer, Nvidia GPU
Software: Android Studio, OpenCV, GitHub
Communication: Design proposals, technical reports, instruction manuals, presentations
Languages: English (fluent), Chinese (fluent), Japanese (beginner)

Experience

Farzaneh Najafi |Atlanta, GA *May – August 2016*
Machine Learning Researcher

- Designed and trained computer vision models for pose estimation using Vision Transformers, achieving 90\% prediction accuracy.
- Built a fully automated data preprocessing and training pipeline to improve experiment reproducibility and eliminate manual intervention.
- Optimized research codebases through algorithmic and GPU-level improvements, achieving up to 10$\times$ faster execution.
- Implemented distributed multi-GPU training using PyTorch DistributedDataParallel (DDP) to accelerate deep learning experiments.
- Collaborated with interdisciplinary researchers to design scalable machine learning workflows for computer vision research.

Projects

Activation Compression Training (ACT) Framework *Semester 2017*

- Developed a deep learning training framework for ultra-low-bit activation compression (2-bit and 1-bit) with minimal accuracy loss.
- Designed frequency-aware weight initialization and spectral noise injection techniques to stabilize training under aggressive quantization.
- Implemented advanced training infrastructure including PyTorch DDP, AMP, EMA, SWA, Mixup, and CutMix.
- Achieved state-of-the-art results on CIFAR-10 using compressed activations with accuracy comparable to full-precision models.

Relevant Coursework

Advanced classes: Machine Learning, GPU Programm:General Computing, Multivariable Calculus,